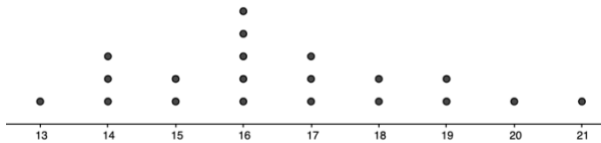


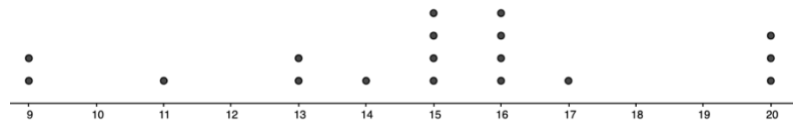
The data shown below are the tip amounts at two different restaurants on the same night.

**Restaurant 1**



**Tip Amount (in dollars)**

**Restaurant 2**



**Tip Amount (in dollars)**

1. The value for the larger median restaurant tips is \$16.

Restaurant 1 → \$16

Restaurant 2 → \$15

2. The value for the larger mean restaurant tips is \$16.55.

Restaurant 1 → \$16.55

Restaurant 2 → \$15

3. Which restaurant has more variability in the tip amounts?

Restaurant 2 ( $20 - 9 = 11$ )

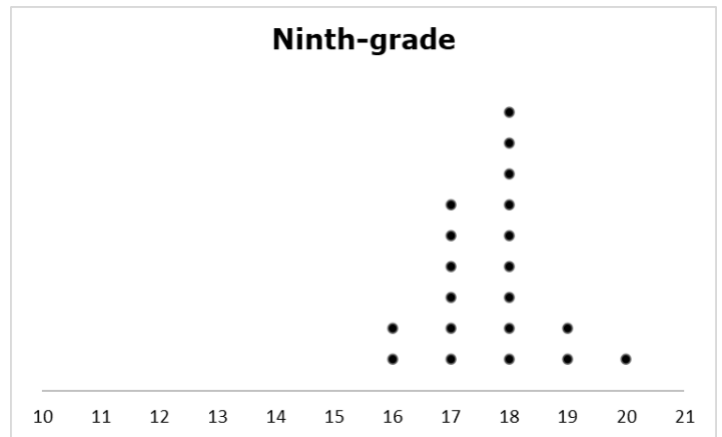
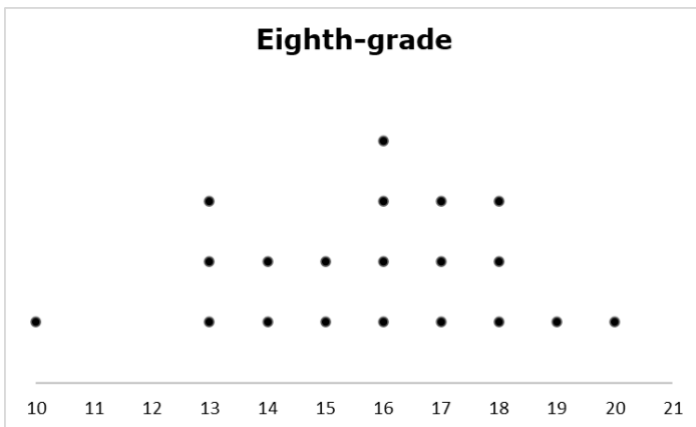
4. Which measure of variability would be best used to describe the graphs? Explain your reasoning.

IQR because you'd be less affected by extremely high and low tips (i.e., outliers)

5. Based on the data, which restaurant made more money in tips?

Restaurant 1

Mrs. Gottberg recorded the time her eighth-grade students, and her ninth-grade students spent studying each week. She recorded the data and then created dot plots to show what she found.



**Time (in hours)**

6. The value of the mean for the larger amount of time spent studying was

17.7.      8<sup>th</sup> → 15.75  
9<sup>th</sup> → 17.7

7. The value of the median for the larger amount of time spent studying was

18.      8<sup>th</sup> → 16  
9<sup>th</sup> → 18

8. Which class has more variability in their studying times?

8<sup>th</sup> ( $20 - 10 = 10$ )

9. Which measure of variability could be used to best describe the graphs? Explain your reasoning.

8<sup>th</sup> → IQR because of outlier at 10  
9<sup>th</sup> → IQR because it's slightly skewed

10. Based on the data, which grade spent the most time studying?

Ninth grade